LOG-CONCAVITY OF BRUHAT INTERVALS IN WEYL GROUPS: EXTENDED VERIFICATION, AN EXTREMAL CONJECTURE, MAHONIAN ASYMPTOTICS, AND EQUALITY CASES

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Abstract. Brenti conjectured that the rank sequence of every Bruhat interval in a finite Weyl group is log-concave; the statement is known to fail for the non-crystallographic Coxeter group H_3. We report an extended computational verification of the conjecture, exhaustively checking 1,079,490,991 Bruhat intervals across the Weyl groups of types A_2–A_7, B_2–B_6, D_4–D_6, E_6, F_4, and G_2 in exact integer arithmetic, with zero violations found; this extends the previously recorded verification frontier by roughly an order of magnitude and closes several cases (A_6, A_7, the full range of B_5, B_6, D_6, E_6) left open in Brenti's own list. We complement this with non-exhaustive evidence in higher rank: near-top slab sweeps in A_8–A_{10}, D_7–D_8, E_7, and 124,944 sampled and seeded intervals in B_7, B_8, D_7, E_7. The data suggest an extremal principle: in every irreducible, simply-laced Weyl group the global minimum of the log-concavity ratio over all Bruhat intervals is attained by the full interval $[e, w_0]$; we show both hypotheses — irreducibility and the simply-laced restriction — are necessary, and record a proved instance of the principle restricted to rationally smooth intervals. In type A this minimum is the central ratio of the Mahonian distribution; we prove an unconditional local limit theorem pinning its value near the center for every m, determine the global minimum exactly, in integer arithmetic, for every $4 \le m \le 560$, and prove the sharp asymptotic $\sigma_m^2(r_m - 1) = 1 - \frac{27}{25}m^{-1} + O(m^{-2})$ conditional on a single explicitly stated core lemma on the exponentially tilted Mahonian distribution — needed only for $m \ge 561$, and itself reduced, with explicit constants throughout, to four explicitly stated open cumulant statements. We also record a classification of the observed equality cases as rank-two dihedral parabolic patterns of order $m \in \{4, 6\}$, together with a candidate explanation, via the crystallographic restriction, for why Weyl groups escape the failure Brenti records for the non-crystallographic group H_3, whose counterexample interval sits on a dihedral core of order $m = 5$.

1. Introduction

Let W be a finite Weyl group with simple reflections S, length function ℓ, and Bruhat order $\le$. For $u \le v$ in W, the *Bruhat interval* $[u, v] = \{z \in W : u \le z \le v\}$ carries a natural graded rank sequence

$$a_k([u, v]) \ = \ \#\{z \in [u, v] : \ell(z) = \ell(u) + k\}, \qquad 0 \le k \le \ell(u, v) := \ell(v) - \ell(u),$$

and $[u, v]$ is *rank log-concave* if $a_k^2 \ge a_{k-1}a_{k+1}$ for every $1 \le k \le \ell(u, v) - 1$. Brenti asked, as Conjecture 2.11 of his 2024 survey of open problems on Coxeter groups and unimodality [1], whether this holds for *every* Bruhat interval of *every* finite Weyl group. The question is delicate: the log-concavity statement is false for general finite Coxeter groups, failing already for the non-crystallographic group H_3 (Section 2), so any positive answer for Weyl groups specifically must use the crystallographic structure in an essential way, not merely finiteness of the group.

At the time of writing, the conjecture was verified only for a modest list of small-rank cases: types A_n and D_n for $n \le 5$, B_n for $n \le 4$, B_5 restricted to intervals of length at least 20, F_4, and the dihedral groups [1]. Beyond direct verification, the strongest structural result in the literature is the

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top-heaviness inequality of Björner and Ekedahl [2] for parabolic lower intervals, which constrains the *shape* of the rank sequence but does not compare ratios $a_k^2/(a_{k-1}a_{k+1})$ across different intervals and so neither implies nor suggests any statement about *where* the log-concavity ratio is smallest.

Contributions. We report four results.

(i) An exhaustive computational verification that pushes the confirmed frontier well past Brenti's list: every Bruhat interval is checked, in exact integer arithmetic, in types A_2–A_7, B_2–B_6 (the *full* range, not only $\ell(u,v) \geq 20$), D_4–D_6, E_6, F_4, and G_2 — 1,079,490,991 intervals total, zero violations (Theorem 4.1). We complement this with non-exhaustive evidence at higher rank: near-top slab sweeps reaching A_{10}, D_8, and E_7 (apparently the first log-concavity data of any kind recorded for E_7), and 124,944 sampled and targeted intervals in B_7, B_8, D_7, E_7.

(ii) An extremal conjecture, apparently new: in every irreducible, simply-laced Weyl group, the global minimum of the log-concavity ratio over *all* Bruhat intervals is attained by the single longest interval $[e, w_0]$ (Conjecture 5.1). This is consistent with every exhaustive and near-top data point we have, including two structural counterexamples we found along the way showing that both hypotheses — irreducibility and the simply-laced restriction — are necessary.

(iii) In type A, a determination — conditional on a single explicitly stated open lemma — of how the minimum log-concavity ratio of the Mahonian distribution (the $[e, w_0]$ interval of S_m) approaches 1, extending the local central-limit-theorem technique of Canfield, Janson, and Zeilberger [3] from the central coefficient of the Gaussian binomial to the global minimum and an explicit constant. A local limit theorem pinning the ratio near the center is unconditionally proved for every m; the minimum itself is determined exactly, in integer arithmetic, for every $4 \leq m \leq 560$; and the sharp asymptotic $\sigma_m^2(r_m - 1) = 1 - \frac{27}{25}m^{-1} + O(m^{-2})$ is proved conditional on one explicitly stated lemma about the exponentially tilted Mahonian distribution (Conjecture 6.3), needed only for $m \geq 561$, with the reduction complete and every constant explicit; the lemma is in turn reduced to four explicitly stated open cumulant statements (Section 6).

(iv) A classification of the observed equality cases — log-concavity holding with equality — as rank-two dihedral parabolic patterns of order $m \in \{4, 6\}$, together with a candidate explanation, via the crystallographic restriction theorem, for why Weyl groups escape the H_3 failure: the H_3 counterexample sits on a dihedral core of order $m = 5$, which is precisely the order excluded from crystallographic reflection groups (Section 7).

Related work. Beyond Brenti's survey and Björner–Ekedahl, several recent papers touch adjacent territory without overlapping our results. Burrull, Gui, and Hu [10] prove an asymptotic log-concavity result for dominant lower intervals in *affine* Weyl groups under dilation, a genuinely different regime (affine, parabolic, asymptotic-in-dilation) from the finite, full-interval statement studied here, and record a non-unimodal example due to Stanton showing the naive parabolic analogue of Conjecture 2.11 can fail. Kessouri, Ahmia, Arslan, and Mesbahi [18] prove ordinary (finite-n) log-concavity of type-B q-Mahonian numbers, with no asymptotic statement, and so do not overlap with our type-A asymptotic (iii). We also note, as a caution about the reliability of open-problem surveys in general rather than as a conflict, that another conjecture from the same survey was recently refuted by Chapelier-Laget and Fromentin [19]. A fresh literature sweep at the time of writing (2026-08-06) found no paper stating, proving, or refuting any of (i)–(iv).

Every claim in this note is accompanied by a reproducible verification artifact: source code, exact-arithmetic result logs, and (where a claim is fully proved) machine-checked numeric certificates are available at [repository URL to be added on submission].

2. Preliminaries

Let W be a finite Coxeter group with simple generators S, length function ℓ, and Bruhat order $\leq$: $u \leq v$ iff some (equivalently, every) reduced word for v contains a reduced word for u as a subword. For a finite *Weyl* group W, i.e. the group generated by reflections in the root system of a Cartan matrix, W is realized concretely as a subgroup of orthogonal transformations of a Euclidean space and every construction below is computed directly from the Cartan matrix (Section 3).

For $u \leq v$ write $[u, v] = \{z \in W : u \leq z \leq v\}$ for the Bruhat interval, graded by $\ell(z) - \ell(u)$, with rank sequence

$$a_k([u, v]) = \#\{z \in [u, v] : \ell(z) = \ell(u) + k\}, \qquad 0 \leq k \leq \ell(u, v) := \ell(v) - \ell(u).$$

All $a_k([u, v]) > 0$ for Bruhat intervals, so we may define the *log-concavity ratio*

$$\rho([u, v]) = \min_{1 \leq k \leq \ell(u,v)-1} \frac{a_k([u, v])^2}{a_{k-1}([u, v])\, a_{k+1}([u, v])} \qquad (\ell(u, v) \geq 2),$$

and $[u, v]$ is rank log-concave exactly when $\rho([u, v]) \geq 1$. Brenti's Conjecture 2.11 [1] asserts $\rho([u, v]) \geq 1$ for every Bruhat interval of every finite *Weyl* group (crucially: not every finite Coxeter group, by Example 2.2 below).

The interval $[e, w_0]$, where w_0 is the longest element, is distinguished: its rank sequence is exactly the coefficient sequence of the Poincaré polynomial $W(q) = \sum_{w \in W} q^{\ell(w)} = \prod_i [d_i]_q$ (product over the degrees d_i of W), so $a_k([e, w_0]) = [q^k]\, W(q)$.

Example 2.1 (Type A: Mahonian numbers)**.** For $W = S_m$ (type A_{m-1}), ℓ is the number of inversions, $W(q) = [m]_q! = \prod_{i=1}^{m}(1 + q + \cdots + q^{i-1})$, and $a_k([e, w_0]) = I_m(k) := \#\{\sigma \in S_m : \mathrm{inv}(\sigma) = k\}$, the classical Mahonian numbers. Log-concavity of $(I_m(k))_k$ itself is classical ([4]; also a consequence of [5, 6] applied to the factorization of $[m]_q!$); our contribution in Section 6 is the asymptotic behaviour of the *minimum ratio*, not log-concavity itself.

Example 2.2 (Failure for general Coxeter groups: H_3)**.** We have independently confirmed the following directly against the primary source. Let (W, S) be of type H_3, $S = \{s_1, s_2, s_3\}$ with $m(s_1, s_2) = 5$ and $m(s_2, s_3) = 3$, and let $u = s_3$, $v = s_1 s_2 s_3 s_2 s_1 s_2 s_1 s_3$. Then [1, Conjecture 2.11] records the rank generating function of $[u, v]$ as

$$1 + 3t + 5t^2 + 7t^3 + 10t^4 + 10t^5 + 5t^6 + t^7,$$

i.e. rank sequence $(a_0, \ldots, a_7) = (1, 3, 5, 7, 10, 10, 5, 1)$. This sequence is unimodal but not log-concave: at $k = 3$, $a_3^2 = 49 < a_2 a_4 = 50$, a margin of -1. We verified this quotation directly against the text of [1] (*Some open problems on Coxeter groups and unimodality*, to appear in Proc. Sympos. Pure Math. **110**; page 10 of the arXiv:2410.09897 PDF, Conjecture 2.11), where it appears verbatim, including the stated Coxeter parameters $m(s_1, s_2) = 5$, $m(s_2, s_3) = 3$; the arithmetic $7^2 = 49 < 5 \cdot 10 = 50$ checks out exactly. The rank-2 parabolic $\langle s_1, s_2 \rangle$ generating the failure is dihedral of order $2m(s_1, s_2) = 10$, i.e. an $I_2(5)$ core — this is the fact exploited in Section 7.

3. Computational methodology

All computations use exact integer (arbitrary-precision) arithmetic; no floating point is involved in any claim labeled a theorem. Three independent implementations were used, cross-checked against each other and against known invariants before any new result was trusted:

- A generic Cartan-matrix engine (`weyl.py`), which builds any finite Weyl group directly from its Cartan matrix and computes $[u, v]$ by bitset-encoded up/down Bruhat-order BFS; used for the exhaustive tier.

- A complement-BFS engine for near-top intervals (`scaled.py`, `scaled_general.py`), which computes rank sequences of lower intervals $[e, v]$ with v close to w_0 by BFS on the (small) complement $W \setminus [e, v]^{\downarrow}$ rather than on $[e, v]$ itself, making groups far too large to enumerate exhaustively tractable in a thin slab below w_0.
- A root-action engine (`fast.py`) for high-throughput sampling and seeded perturbation search.

Each engine was validated against independent invariants before use: known group order $|W|$; length $\equiv$ number of inversions (root-system definition) for every enumerated element; Poincaré polynomial $\equiv$ the degree-product formula $\prod_i [d_i]_q$; and, where two engines cover the same group, engine-versus-engine agreement on every interval.

We report results in three tiers, and state precisely what each does and does not establish:

Exhaustive: Every Bruhat interval of the group is enumerated and checked. This is the only tier that constitutes a full verification of Conjecture 2.11 for the group in question (Theorem 4.1, Table 1).

Near-top slab: Only lower intervals $[e, v]$ with $\ell(w_0) - \ell(v)$ below a fixed bound are checked; this is evidence for the conjecture and for Conjecture 5.1 in the checked groups, not a full verification.

Sampling / seeded: Random or targeted samples; coverage evidence only.

Remark 3.1 (E_6/B_6 segmentation and a lost witness). The E_6 and B_6 exhaustive runs were each executed as two complementary, non-overlapping segments (partitioned by the bottom element u) after an earlier single-pass attempt was interrupted by a process kill; the segment boundaries are documented and their union is a genuine full cover of the group (verified: the segment interval counts sum exactly to the totals in Table 1). For B_6 this recovery is additionally corroborated by an independent non-segmented full run, so no data is in question. For E_6, the first segment ($u < 6000$) survived only as a running-minimum checkpoint stream, without the interval $[u, v]$ or rank sequence attaining that minimum; we therefore report the exact minimum ratio for E_6 but cannot exhibit its witness interval in this note, and flag a re-scan of $u < 6000$ as a documented, low-cost follow-up rather than a gap that affects the truth of Theorem 4.1.

All raw result logs, per-run dossiers, and the CI configuration used to reproduce the exhaustive tier are included in the accompanying repository ([repository URL to be added on submission]).

4. Verification results

Theorem 4.1 (Exhaustive tier). *Let W be one of the Weyl groups $A_2, \dots, A_7$, $B_2, \dots, B_6$, D_4, D_5, D_6, E_6, F_4, G_2. Then every Bruhat interval $[u, v] \subseteq W$ is rank log-concave. The verification is exhaustive: every interval of length ≥ 2 was enumerated and checked in exact integer arithmetic (Table 1), totalling* 1,079,490,991 *intervals with zero violations.*

Remark 4.2. Relative to the previously recorded frontier (Brenti's list: A_n, D_n for $n \leq 5$; B_n for $n \leq 4$; B_5 only for $\ell(u, v) \geq 20$; F_4; dihedral groups), the cases A_6, A_7, B_5 (full range), B_6, D_6, E_6 are new.

Theorem 4.3 (Near-top tier — lower intervals only; not exhaustive). *For each pair (W, c) below, every lower interval $[e, v]$ with $\ell(w_0) - \ell(v) \leq c$ is rank log-concave; moreover the minimum of $\rho([e, v])$ over this slab is attained at $v = w_0$ (Table 2):*

$$(A_7, 4),\ (A_8, 4),\ (A_9, 3),\ (D_7, 3),\ (D_8, 3),\ (E_7, 2).$$

In A_{10}, $[e, w_0]$ and a partial slab (11 of 65 candidates at $c = 2$) were checked, all log-concave, with one proper interval exactly tying $\rho([e, w_0])$.

TABLE 1. Exhaustive tier: every Bruhat interval, exact integer arithmetic.

Group	$\lvert W\rvert$	#intervals ($\ell \geq 2$)	min ratio	witness (index k)
A_2	6	19	2.000000	$[e, 121]$, $k = 1$
A_3	24	213	1.388889	$[e, 12321]$, $k = 2$
A_4	120	3,781	1.210000	$[e, w_0]$, $k = 5$
A_5	720	98,407	1.122222	proper (ties $[e, w_0]$), $k = 7$
A_6	5,040	3,550,919	1.079096	proper (ties $[e, w_0]$), $k = 10$
A_7	40,320	170,288,585	1.054250 = 919681/872356	proper, $\ell = 27$ (ties $[e, w_0]$), $k = 14$
B_2	8	33	1.000000	$[e, 1212]$, $k = 2$
B_3	48	847	1.000000	$[e, 2323]$, $k = 2$
B_4	384	40,249	1.000000	$[e, 3434]$, $k = 2$
B_5	3,840	3,089,459	1.000000	$[e, 4545]$, $k = 2$
B_6	46,080	350,676,009	1.000000	$[e, 5656]$, $k = 2$
D_4	192	9,817	1.136232	proper (ties $[e, w_0]$), $\ell = 11$, $k = 5$
D_5	1,920	745,377	1.069459	proper (ties $[e, w_0]$), $\ell = 19$, $k = 10$
D_6	23,040	84,339,681	1.040703	proper, $\ell = 27$ (ties $[e, w_0]$), $k = 15$
E_6	51,840	466,250,713	1.028446	witness unrecoverable, see Remark 3.1
F_4	1,152	396,809	1.000000	$[e, 2323]$, $k = 2$
G_2	12	73	1.000000	$[e, 1212]$, $k = 2$

TABLE 2. Near-top slab tier: lower intervals $[e, v]$, $\ell(w_0) - \ell(v) \leq c$ (not exhaustive). Exact fractions for the min ratio, where computed, are given in the text.

Group	$\lvert W\rvert$	slab	#checked	min ratio	at k
A_7	40,320	$c \leq 4$	full	1.054250	$[e, w_0]$, 14
A_8	362,880	$c \leq 4$	full	1.038942	$[e, w_0]$, 18
A_9	3,628,800	$c \leq 3$	209/209	1.028950	$[e, w_0]$, 22
A_{10}	39,916,800	$c \leq 2$	11/65*	1.022102	tie†, 27
D_7	322,560	$c \leq 3$	112/112	1.025574	$[e, w_0]$, 21
D_8	5,160,960	$c \leq 3$	156/156	1.017122	$[e, w_0]$, 28
E_7	2,903,040	$c \leq 2$	full	1.011829	$[e, w_0]$, 31

*partial, resumable; †$[e, w_0]$ attains the minimum and one proper interval exactly ties it.

Remark 4.4. The two exact rational minima recorded in Table 2: $\rho([e, w_0]) = 919681/872356$ in A_7, and $\rho([e, w_0]) = 65523/64757$ in E_7; the latter's rank sequence sums to exactly $2{,}903{,}040 = |W(E_7)|$, confirming it is the full interval and not a shorter proper one.

Remark 4.5. Theorem 4.3 checks only lower intervals in a thin slab below w_0; it is evidence for Conjecture 2.11 and for Conjecture 5.1 in these groups, not a verification of either beyond the stated slab. E_7 is, to our knowledge, the first log-concavity verification data of any kind recorded for that group.

Remark 4.6. The seeded probe attempts a fixed number of perturbations of the $m = 4$ dihedral core (the header count of each seeded run above: 4,000; 100,000; 100,000); an attempt is discarded, uncounted, whenever the targeted perturbation does not realize a genuine Bruhat up-cover chain. The counts reported in the #intervals column and used throughout this paper are the surviving, actually-checked intervals only.

Proposition 4.7 (Sampling tier — not exhaustive; coverage statement only)**.** (i) 60,000 *Bruhat intervals (*20,000 *each in B_7, D_7, E_7), sampled uniformly from a random-walk ensemble with* $4 \leq \ell(u, v) \leq 12$ *(seed 7, reproducible), are all rank log-concave.* (ii) 64,944 *seeded intervals*

Table 3. Sampling / seeded tier (not exhaustive; coverage counts).

Probe	Group	#intervals	design	min ratio	note
random (seed 7)	B_7	20,000	$4 \le \ell(u,v) \le 12$	1.000000	$(1,2,2,2,1)$ pattern
random (seed 7)	D_7	20,000	same	$1.157143 = 81/70$	proper, k=4
random (seed 7)	E_7	20,000	same	$1.142400 = 14641/12816$	proper, k=5
seeded (seed 3)	B_7	1,324	m=4 core + 0–6 covers	1.000000 only at pure core	28 ratio-1 hits
seeded (seed 4)	B_7	32,458	m=4 core + 0–10 covers	pert.≥ 1: ≥ 1.079153	margins 4–8
seeded (seed 4)	B_8	31,162	m=4 core + 0–10 covers	pert.≥ 1: ≥ 1.087990	margins 4–8

containing an $m = 4$ dihedral braid core perturbed by 0–10 extra cover steps are all rank log-concave; every perturbation that destroys the pure dihedral pattern strictly raises the ratio (Table 3; the seeded search attempted more perturbations than this, discarding any that did not realize a genuine cover chain — see the remark following the table).

In total, the sampling and seeded probes cover 124,944 intervals with zero violations and zero equality cases outside the $(1, 2, \ldots, 2, 1)$ pattern classified in Section 7.

5. An extremal conjecture

Conjecture 5.1 (F1). *Let W be a finite Weyl group that is irreducible and simply-laced (type A_n, D_n, E_6, E_7, or E_8), with longest element w_0. Then*

$$\min_{\substack{[u,v]\subseteq W\\ \ell(u,v)\ge 2}} \rho([u,v]) \;=\; \rho([e,w_0]) \;=\; \min_{1\le k\le \ell(w_0)-1} \frac{b_k^2}{b_{k-1}b_{k+1}},$$

where $b_k = [q^k]\,W(q)$: the global minimum of the log-concavity ratio over all Bruhat intervals of W is attained by the full interval $[e, w_0]$, at a central index. Proper intervals may tie this minimum (observed for A_5, A_6, A_7, D_6 in the exhaustive tier, and for a further proper interval in the A_{10} near-top slab) but never fall below it.

Both hypotheses in Conjecture 5.1 are necessary; we record the two sharp counterexamples that force this wording.

Example 5.2 (Simply-laced is necessary)**.** In B_3, the interval $[e, (s_2s_3)^2]$ has rank sequence $(1, 2, 2, 2, 1)$ and $\rho = 2^2/(2 \cdot 2) = 1$ exactly, on a plateau, strictly below $\rho([e, w_0(B_3)]) = 8/7$. The analogous B_4 interval $[e, (s_3s_4)^2]$ gives $\rho = 1 < 968/897 = \rho([e, w_0(B_4)])$. Note $[e, (s_2s_3)^2]$ is *rationally smooth* (its rank sequence is palindromic), so dropping the simply-laced hypothesis falsifies the conjecture even within the rationally-smooth subclass discussed below.

Example 5.3 (Irreducible is necessary — apparently new)**.** Let $W = A_1 \times D_4$ (simply-laced, reducible) and $v = (e, w_0(D_4)) \in W$. The interval $[e, v]$ is isomorphic, as a graded poset, to $[e, w_0(D_4)]$ inside the D_4 factor, with rank sequence $P_{D_4} = (1, 4, 9, 16, 23, 28, 30, 28, 23, 16, 9, 4, 1)$ and $\rho([e, v]) = 28^2/(23 \cdot 30) = 392/345 = 1.1362318\ldots$ at $k = 5$. This is a *proper* interval of W: $\ell(v) = 12 < 13 = \ell(w_0(W))$. The full interval has rank sequence $[2]_q \cdot P_{D_4} = (1, 5, 13, 25, 39, 51, 58, 58, 51, 39, 25, 13, 5, 1)$ and $\rho([e, w_0(W)]) = 58^2/(51{\cdot}58) = 58/51 = 1.1372549\ldots$ Since $392 \cdot 51 = 19992 < 20010 = 345 \cdot 58$,

$$\rho([e,v]) \;=\; \frac{392}{345} \;<\; \frac{58}{51} \;=\; \rho([e,w_0(W)]), \qquad \text{gap } \frac{58}{51} - \frac{392}{345} = \frac{2}{1955} \approx 0.00102,$$

so a *proper* Bruhat interval of the reducible simply-laced group $A_1 \times D_4$ strictly beats $[e, w_0]$: the extremal principle fails outright once reducibility is allowed, not merely on some restricted subclass. (The witness v happens to be smooth, but this already refutes the unrestricted statement for reducible W, independently of smoothness.) An exhaustive check of all 27 reducible simply-laced types of rank ≤ 6 found $A_1 \times D_4$ to be the *only* violator.

Remark 5.4 (Status: conjectural, not proved). Conjecture 5.1 itself — for arbitrary $[u, v]$, not merely rationally smooth ones — is not proved for any infinite family. What supports it is: exhaustive verification with zero violations for every irreducible simply-laced group in the exhaustive tier (A_2–A_7, D_4–D_6, E_6; Table 1), in every case with $[e, w_0]$ attaining (and, in A_5, A_6, A_7, D_6, exactly tied by a specific proper interval); and near-top slab confirmation with no exception in A_8–A_{10}, D_7–D_8, E_7 (Table 2). Neither tier constitutes a proof for general rank.

Theorem 5.5 (F1 restricted to rationally smooth intervals)**.** *Let W be a finite irreducible simply-laced Weyl group of rank ≤ 6 ($A_1, \ldots, A_6, D_4, D_5, D_6, E_6$). For every rationally smooth $v \in W$ — i.e. every v for which the Poincaré polynomial $P_{[e,v]}(q)$ factors as a product of q-integers* [11] *— $\rho([e, v]) \geq \rho([e, w_0])$, with equality iff $v = w_0$. Unconditionally, this also holds for every m in type A_{m-1} with $m \leq 17$, modulo the classical rational-smoothness factorization fact cited above, which our own working notes verified computationally only through $m \leq 7$ rather than re-deriving in full generality; we flag this explicitly rather than presenting the $m \leq 17$ range as resting on nothing beyond our own work.*

Conjecture 5.6 (Staircase domination)**.** *In type A_{m-1}, the exponent multiset of the rationally-smooth Poincaré polynomial P_v is always a gap-free ("staircase") multiset dominated (coefficient-wise, after sorting) by the degree multiset $\{2, \ldots, m\}$.*

Remark 5.7. Conjecture 5.6 is verified without exception on 91,355 dominated pairs, and, in the specific instances Theorem 5.5 needs, on a further 131,036 pairs for $m \leq 17$; it has not been proved for general m. Granting it, the $m \leq 17$ restriction in Theorem 5.5's type-A clause can be removed, giving the same inequality for every m.

Remark 5.8. Theorem 5.5 is a genuine partial proof, but of a *strictly weaker* statement than Conjecture 5.1: it compares $\rho([e, v])$ against $\rho([e, w_0])$ only for v rationally smooth, not for arbitrary v. Both Examples 5.2 and 5.3 witness this restricted statement's own sharpness (the B_3 counterexample is itself rationally smooth), so the same irreducible-simply-laced hypotheses are necessary here too. Type D_n for $n \geq 7$ and types E_7, E_8 remain open even for this restricted subclass. This theorem is the paper's best current route to a proved instance of the extremal principle, and we record it as the natural target for future work on Conjecture 5.1 in full.

6. The type-A case: Mahonian asymptotics

Throughout this section $W = S_m$ (type A_{m-1}), $I_m(k) = \#\{\sigma \in S_m : \mathrm{inv}(\sigma) = k\}$ so that $a_k([e, w_0]) = I_m(k)$, $N = \binom{m}{2}$, $\sigma^2 = \mathrm{Var}(\mathrm{inv}) = m(m-1)(2m+5)/72$, $r_m(k) = I_m(k)^2/(I_m(k-1)I_m(k+1))$, and $r_m = \min_{1 \leq k \leq N-1} r_m(k)$. Write $y = (k - N/2)/\sigma$, and let $B_m := (S_4 - m)/(240\sigma^4)$ where $S_r = \sum_{j=1}^{m} j^r$, so $B_m = \frac{27}{25} m^{-1}(1 + O(m^{-1}))$.

What is proved unconditionally. The following local limit theorem for the log-concavity ratio, in the classical style of [13], pinning it near the center for *every* m, is fully proved — with every constant explicit and both an analytic argument and an exact finite check covering every case, and independently referee-audited (mathematics and numerics separately, in two independent passes), with the referees' verdict "survives with minor repairs" and those repairs applied.

Theorem 6.1 (Pointwise and window law for the Mahonian ratio)**.** *Let $p(k) = I_m(k)/m!$, $Z(y) = (2\pi\sigma^2)^{-1/2} e^{-y^2/2}$, and let He_4 denote the fourth probabilists' Hermite polynomial.*

(a) (Pointwise.) *For every $m \geq 4$ and every $0 \leq k \leq N$,*

$$\left| p(k) - Z(y)\left[1 - \tfrac{B_m}{12}\mathrm{He}_4(y)\right] \right| \;\leq\; \frac{0.45}{\sigma m^2}.$$

(b) (Window ratio law.) *For every fixed $y_0 \geq 0$ there are explicit $C_2(y_0)$ and $m_1(y_0)$ (e.g. $C_2(1) = 3.1$, $m_1(1) = 180$; the constant grows with y_0, e.g. $C_2(3) = 3940$) such that for every $m \geq m_1(y_0)$ and every k with $|y| \leq y_0$,*

$$\sigma^2 \log r_m(k) \;=\; 1 + B_m(y^2 - 1) + E_1(k), \qquad |E_1(k)| \leq \frac{C_2(y_0)}{m^2}.$$

Moreover (a) holds, with the same displayed constant, for every $4 \leq m \leq 109$ by direct exact-arithmetic verification, and the window law of (b) holds, with the same displayed constants, for every $4 \leq m \leq 229$; both are also fully covered by the analytic argument above outside these ranges, so neither part carries a threshold in m — the exact-arithmetic ranges are an independent check, not the limit of what is proved, and the window law in (b) is unconditional (no asymptotic-only asterisk) throughout the range any downstream use in this paper requires.

Theorem 6.1 determines the central ratio exactly to the stated order: at $y = 0$, part (b) gives

$$\sigma^2\bigl(r_m(\lfloor N/2 \rfloor) - 1\bigr) \;=\; 1 - \tfrac{27}{25}m^{-1} + O(m^{-2}), \tag{1}$$

which we have independently confirmed to four significant digits against exact Mahonian data by $m = 50$.

The second unconditional ingredient is exact: the global minimum itself — its location, its value, and the sharp finite-m bound — has been computed exactly for every $4 \leq m \leq 560$.

Theorem 6.2 (Exact finite range)**.** *For every $5 \leq m \leq 560$, with every verdict computed in exact integer or rational arithmetic:*

(i) *the minimum of $r_m(k)$ over $1 \leq k \leq N - 1$ is attained at the central index $k = \lfloor N/2 \rfloor$ (for N odd, the mirror index $\lceil N/2 \rceil$ ties it exactly, as palindromy forces);*
(ii) *consequently $r_m = r_m(\lfloor N/2 \rfloor)$;*
(iii) *$\sigma^2(r_m - 1) \geq \frac{187}{216}$, with equality if and only if $m = 6$;*
(iv) *$m \mapsto \sigma^2(r_m - 1)$ is strictly increasing for $6 \leq m \leq 560$, reaching $\sigma^2(r_m - 1) = 0.998072\ldots$ at $m = 560$, where $m\bigl(1 - \sigma^2(r_m - 1)\bigr) \approx 1.07935$, consistent with the limit $\frac{27}{25} = 1.08$.*

*At $m = 4$, and only there in the computed range, the minimizing index is not central (*argmin $= 2$*, $N/2 = 3$) and $\sigma^2(r_4 - 1) = \frac{91}{108} < \frac{187}{216}$.*

The computation is a checkpointed harness over the q-factorial recurrence for I_m; every verdict is an integer or exact-rational comparison, floating point appears only in printed displays, and the complete result logs — one row for every $4 \leq m \leq 560$, all passing — are archived in the repository.

From the center to the global minimum: the reduction. Equation (1) pins the ratio at and near the center, and Theorem 6.2 settles every $4 \leq m \leq 560$ outright. What remains for the sharp asymptotic is to rule out the ratio dipping below its central value somewhere in the bulk for $m \geq 561$. This has now been carried out in full, with every constant explicit, *modulo exactly one statement*: an estimate for the exponentially tilted Mahonian distribution in the deep-tilt regime, stated precisely as Conjecture 6.3 below. We first fix the frame.

For $\lambda \in \mathbb{R}$ let $U_1^\lambda, \ldots, U_m^\lambda$ be independent, with U_j^λ supported on $\{0, 1, \ldots, j-1\}$ and $\Pr(U_j^\lambda = i) \propto e^{-\lambda i}$, and set $S_\lambda := \sum_{j=1}^m U_j^\lambda$ (not to be confused with the power sums S_r above), $\mu(\lambda) := \mathbb{E} S_\lambda$. At $\lambda = 0$ this is the inversion statistic: $\Pr(S_0 = k) = I_m(k)/m!$. Since $\mu'(\lambda) = -\operatorname{Var} S_\lambda < 0$, every interior k ($1 \leq k \leq N - 1$) has a unique *mean-matching tilt* $\lambda(k)$ with $\mu(\lambda(k)) = k$; write $s^2 = s^2(k) := \operatorname{Var} S_{\lambda(k)}$, let κ_3, κ_4 denote the third and fourth cumulants of $S_{\lambda(k)}$, and let $\varphi(t) := \mathbb{E}\, e^{it(S_{\lambda(k)} - k)}$. By palindromy $r_m(N - k) = r_m(k)$ and $\lambda(N - k) = -\lambda(k)$, so we take $k \leq N/2$, i.e. $\lambda(k) \geq 0$, throughout, and set $\omega := m\lambda(k)$.

Conjecture 6.3 (Core lemma CL(79, 20, 0.89))**.** *For every $m \geq 561$ and every interior k with $s^2(k) \geq 79$ and $4/m < \lambda(k) \leq 0.89$,*

$$r_m(k) - 1 \;=\; \frac{1}{s^2}\Big(1 + \theta\,\frac{20}{\min(m, s^2)}\Big) \qquad \text{for some } \theta = \theta(m,k) \in [-1,1].$$

Remark 6.4. Three notes on the statement. (a) The hypothesis $s^2 \geq 79$ is part of the frozen specification of the reduction but never binds: on the stated tilt range $s^2 \geq 1122800/7921 > 141$ is proved. (b) Only the lower-bound half ($\theta \geq -1$) is consumed by Theorem 6.5. (c) The complementary small-tilt range $\lambda(k) \leq 4/m$ is not part of the conjecture because it is covered unconditionally by the proved window law (region (d) of Remark 6.6). The reduction behind Theorem 6.5 needs this statement for every $m \geq 401$; the range $401 \leq m \leq 560$ is discharged by the exact computation of Theorem 6.2, which is why the conjecture is stated, and is needed, only for $m \geq 561$.

Theorem 6.5 (F2, sharp asymptotic — conditional)**.** *Assume Conjecture 6.3. Then*

$$\sigma^2(r_m - 1) \;=\; 1 - \tfrac{27}{25}m^{-1} + O(m^{-2}), \qquad \text{equivalently} \qquad r_m - 1 \sim \frac{36}{m^3},$$

and in particular $\sigma^2(r_m - 1) \to 1$. The error term is explicit and two-sided: for every $m \geq 561$,

$$1 - B_m - \frac{C_A}{m^2} \;\leq\; \sigma^2(r_m - 1) \;\leq\; 1 - B_m + \frac{1.8}{m^2}, \qquad C_A = 37997.85,$$

the upper bound holding unconditionally for every $m \geq 180$.

Remark 6.6 (Shape of the reduction, and where the conjecture enters)*.* The proof is an assembly in which every constant is explicit and every step is archived, with machine-verified certificates, in the repository; we record its shape here. Take $m \geq 561$ and (WLOG) $k \leq N/2$. The interior indices split into four regions by the size of k and of $\omega(k) = m\lambda(k)$, with $K_c := \min(\frac{7}{10}m,\, m-1)$ for $m < 1581$ and $K_c := m - 1$ beyond:

(a) $k = 1$: an exact pentagonal-recurrence bound gives $r_m(1) - 1 \geq \frac{m-1}{2(m+1)}$, hence $\sigma^2(r_m(1) - 1) \geq 10^5$;
(b) $2 \leq k \leq K_c$: an elementary two-term bound gives $r_m(k) - 1 \geq \frac{m-1}{2k(m+k)}$, hence $\sigma^2(r_m(k) - 1) \geq 1879$;
(c) $k > K_c$ with $\omega(k) > 4$ (the deep-tilt region): here $\min(m, s^2) \geq 79.5$, $s^2 \leq 0.72711\,\sigma^2$, and $\lambda(k) \leq \log(17/7) < 0.89$, all proved; Conjecture 6.3 — consumed here and nowhere else — gives $\sigma^2(r_m(k) - 1) \geq \big(1 - \frac{20}{79.5}\big)/0.72711 \geq 1.0293$;
(d) $k > K_c$ with $\omega(k) \leq 4$: the window law of Theorem 6.1, extended unconditionally and with explicit constants to the full small-tilt window $|\omega| \leq 4$, gives $\sigma^2(r_m(k) - 1) \geq 1 - B_m - C_A/m^2$.

Regions (a)–(c) all exceed $1.02 > 1 - B_m$, so the global minimum lies in region (d); the lower bound follows, and the upper bound is $r_m \leq r_m(\lfloor N/2 \rfloor)$ together with the explicit form of (1). The size of C_A is an artifact of one lossy triangle-inequality step (the measured analogue of the dominant contribution is about 5); any fixed constant suffices for the sharp asymptotic.

Reduction of the core lemma to four cumulant statements. Conjecture 6.3 has itself been reduced, with explicit constants throughout, to four statements about the cumulants of the tilted law. Partition the deep-tilt values $\omega = m\lambda \in (4,\, 0.89\,m]$ into seven bands

$$\mathcal{W}_1 = (4,5],\ \mathcal{W}_2 = (5,6],\ \mathcal{W}_3 = (6,8],\ \mathcal{W}_4 = (8,10],\ \mathcal{W}_5 = (10,20],\ \mathcal{W}_6 = (20,40],\ \mathcal{W}_7 = (40,\infty),$$

each intersected with $\{\lambda \leq 0.89\}$, and normalize the cumulants on the tilt scale:

$$\tilde{\kappa}_3 := \frac{|\kappa_3|\,\lambda}{s^2}, \qquad \tilde{\kappa}_4 := \frac{\kappa_4\,\lambda^2}{s^2}.$$

TABLE 4. Per-band constants in Conjecture 6.7 (bands of $\omega = m\lambda$). The J_0 entries are six-digit displays of exact rational constants archived in the repository.

band ω-range	$\mathcal{W}_1$ $(4,5]$	$\mathcal{W}_2$ $(5,6]$	$\mathcal{W}_3$ $(6,8]$	$\mathcal{W}_4$ $(8,10]$	$\mathcal{W}_5$ $(10,20]$	$\mathcal{W}_6$ $(20,40]$	$\mathcal{W}_7$ $(40,\infty)$
R_3	1.0	1.2	1.5	1.7	2.0	2.1	2.2
R_4	0.8	1.4	2.6	3.5	5.2	6.0	6.6
C_5	0.05	0.06	0.08	0.10	0.15	0.25	0.80
J_0	0.682942	1.10268	1.91562	2.53645	3.66793	4.17806	4.59597

Conjecture 6.7 (The four open statements)**.** *For every $m \geq 561$ and every interior k with $\lambda = \lambda(k) \in (4/m,\, 0.89]$, with $\mathcal{W}$ the band containing $\omega = m\lambda$ and constants R_3, R_4, C_5, J_0 as in Table 4:*

(S1) (banded cumulant scales) $\tilde{\kappa}_3 \leq R_3(\mathcal{W})$ *and* $\tilde{\kappa}_4 \leq R_4(\mathcal{W})$*;*
(S2) (fifth-order core remainder) for $0 \leq t \leq \lambda/2$*, with the continuous branch of* $\log\varphi$ *starting at* $\log\varphi(0) = 0$*,*

$$\left|\log\varphi(t) + \frac{s^2t^2}{2} + \frac{i\kappa_3 t^3}{6} - \frac{\kappa_4 t^4}{24}\right| \;\leq\; C_5(\mathcal{W})\,\frac{s^2t^5}{\lambda^3};$$

(S3) (joint cancellation) $\tilde{\kappa}_3^2 - \tilde{\kappa}_4/2 \leq J_0(\mathcal{W})$*;*
(S4) (a-priori ratio seed) $\left|s^2\big(r_m(k) - 1\big) - 1\right| \leq 0.89$*.*

Proposition 6.8 (Reduction of the core lemma)**.** *Statements (S1)–(S4) of Conjecture 6.7 imply Conjecture 6.3; the composed chain in fact delivers the constant 18.23 in place of 20 for every $m \geq 561$.*

The assembly behind Proposition 6.8 proceeds band by band over Table 4: (S1) and (S3) price the main Gaussian-model term, (S2) prices the core remainder, and (S4) seeds a self-consistency step which then closes with strict contraction; every other ingredient — variance floors and caps, the far-region decay of φ, the crossover and mid-range bounds — is proved with explicit constants. The complete chain, together with an end-to-end machine re-verification of its composed constant (a mixture of exact rational certificates and high-precision numerical evaluation), is archived in the repository.

Remark 6.9 (Status and evidence for (S1)–(S4))*.* All four statements are open; none has a proof, and all four are load-bearing — the assembly consumes each of them, and removing any one leaves no theorem. The list has also moved in both directions under scrutiny: two earlier hypotheses at this level were proved outright and retired, while (S3) and (S4) entered the list when simpler plans were found to fail. We therefore report the numerical evidence with its margins rather than with optimism.

- (S1): the measured suprema of $\tilde{\kappa}_3$ and $\tilde{\kappa}_4$ at the deep corner of $\mathcal{W}_7$ are 2.1215 and 6.3552 (large-m limits 2.1303 and 6.4113) against the caps 2.2 and 6.6 — headroom of only 3.7% and 3.9%, the thinnest margins anywhere in the reduction.
- (S2): measured remainder ratios lie between 0.0083 and 0.2104; the slack over C_5 ranges from a factor 1.6 to a factor 8 depending on the band (the lowest band's slack was measured at interior probe points, not uniformly to its edge).
- (S3): worst measured margin 32.6%, at $(m,\omega) = (561, 5.0)$. This statement cannot be dispensed with: there is a point consistent with (S1) and with $\kappa_4 \geq 0$ at which the estimate (S3) feeds fails by 43%, so a joint bound of this shape is genuinely required — an earlier plan that priced the two cumulants separately is provably dead.

- (S4): a weak a-priori form of the conclusion itself (constant 0.89 where Conjecture 6.3 concludes $20/\min(m, s^2) \le 0.036$, since $s^2 \ge m$ on this range); it is listed separately precisely because assuming it silently would overstate what is proved. Exact verification of the inequality of Conjecture 6.3 at $m = 401$ and $m = 402$ (below its stated range), on 260 adversarially chosen interior indices, passes with a 17× margin, and every tilted ratio measured in this project sits far inside 0.89.

Remark 6.10 (What is claimed, and what is not). Unconditional: Theorems 6.1 and 6.2, and the upper-bound half of Theorem 6.5. Conditional: the sharp asymptotic of Theorem 6.5, on Conjecture 6.3 and on nothing else; and Conjecture 6.3 itself would follow from (S1)–(S4) by Proposition 6.8. Open: statements (S1)–(S4). We do not regard the margins of Remark 6.9 — one of which is under 4% — as grounds for treating any of the four as close to proved; they are open mathematics.

Remark 6.11 (Location and explicit constant). Beyond the sharp order of Theorem 6.5, two refinements. The minimizing index is exactly central for every $5 \le m \le 560$ (Theorem 6.2); for $m \ge 561$, conditional on Conjecture 6.3, the reduction localizes it to the small-tilt window $|\lambda(k)| \le 4/m$ around the center (region (d) of Remark 6.6). Sharpening this to $|\mathrm{argmin} - N/2| \le 1$ for every m is a further, separately open combinatorial statement, equivalent to a third-difference positivity property of $(I_m(k))_k$, verified without exception for $5 \le m \le 56$ but with no proof strategy identified. The explicit finite-m bound $r_m \ge 1 + \frac{187}{216}\sigma_m^{-2}$ for all $m \ge 5$ (sharp at $m = 6$; a corrected value — the naive guess 7/8 fails already at $m = 6$) is exact for $5 \le m \le 560$ (Theorem 6.2(iii)) and holds for every $m \ge 561$ conditional on Conjecture 6.3, since the explicit lower bound of Theorem 6.5 exceeds $\frac{187}{216}$ for every $m \ge 537$; granting the conjecture, the sharp bound therefore holds for every $m \ge 5$, with equality only at $m = 6$.

7. Equality cases

Observation 7.1 (F3, equality classification)**.** Let W be a finite Weyl group and $[u, v] \subseteq W$ a Bruhat interval with $\ell(u, v) \ge 2$ in which $a_k^2 = a_{k-1}a_{k+1}$ for some interior k. Then, in every case observed in our exhaustive and seeded-perturbation data, the full rank sequence of $[u, v]$ is

$$(a_0, \dots, a_{\ell(u,v)}) \;=\; (1, 2, 2, \dots, 2, 1) \qquad (m-1 \text{ interior 2's},\; m \ge 4),$$

i.e. $[u, v]$ is poset-isomorphic to the full Bruhat interval of the dihedral group $I_2(m)$, arising from a rank-two dihedral standard parabolic of braid order m. Since the crystallographic restriction confines rank-two parabolics of Weyl groups to $m \in \{2, 3, 4, 6\}$ (equality requires $m \ge 4$, so effectively $m \in \{4, 6\}$), the realizable equality cores are $m = 4$ (type B_2/C_2, present in B_n and F_4) and $m = 6$ (type G_2); in particular no equality occurs in simply-laced types, where rank-two parabolics have $m \le 3$.

We state this as an empirical observation, not a theorem or even a fully scoped conjecture: it holds without exception in every exhaustively checked group (Table 1; witnesses e.g. B_2 '1212', B_3 '2323', B_4 '3434', B_5 '4545', F_4 '2323', G_2 '1212') and in the 64,944-interval seeded perturbation probe (Table 3), where the equality wall is strict: every perturbation of an $m = 4$ dihedral core by extra cover steps that breaks the pure pattern strictly raises the ratio, with closest non-equality margins of 4–8% in the cases checked. We have not attempted a proof, and have not fully resolved whether the correct scope is "the whole rank sequence is $(1, 2, \dots, 2, 1)$" (the strong form the data support wherever a full witness was recorded) or a weaker, purely local statement about the equality index itself; a route to a genuinely proved short-interval instance exists via the classification of short Bruhat intervals [12], which we have not carried out.

Why Weyl groups escape the H_3 failure. We offer the following as a candidate mechanism, not a theorem. The H_3 counterexample of Example 2.2 sits on an $m = 5$ dihedral core: the parabolic $\langle s_1, s_2\rangle \cong I_2(5)$ is a subsystem of the failing interval's defining data. The crystallographic restriction excludes exactly $m = 5$ (and every $m \notin \{2, 3, 4, 6\}$) from occurring as a rank-two parabolic of any Weyl group. So the only equality/violation-producing dihedral cores available inside a Weyl group are $m = 4$ and $m = 6$, and by Observation 7.1 together with the strict perturbation wall, these sit at ratio exactly 1 and are never pushed below it by any perturbation we have tested. This is consistent with, and offered as an explanation of, why every Weyl group in our exhaustive and near-top data avoids the H_3-type failure, but it is a mechanism sketch resting on an empirical observation (Observation 7.1), not a proof that no Weyl-group interval can ever violate log-concavity.

8. Discussion and open problems

We record what we regard as the most promising directions, roughly in order of tractability.

(1) **Prove Conjecture 6.3 — that is, statements (S1)–(S4) of Conjecture 6.7.** By Theorem 6.5 and Proposition 6.8, these four cumulant statements about the tilted Mahonian law are now the *entire* distance between the sharp asymptotic and an unconditional theorem: the far-region decay, the crossover and mid-range bounds, the variance floors and caps, the assembly itself, and the whole range $4 \leq m \leq 560$ are all in place with explicit constants. The measured margins (Remark 6.9) locate the risk: the cumulant caps of (S1) in the deepest band have under 4% headroom, and that is where a proof attempt — or a counterexample search — should begin.

(2) **Extend Theorem 5.5 past rank 6**, and past type A past $m = 17$ unconditionally by proving Conjecture 5.6. D_n for $n \geq 7$ and E_7, E_8 are open even for the rationally-smooth restriction.

(3) **Self-dual and rank-symmetric Bruhat intervals** ([17], which classifies self-dual intervals) are a natural place to look for further explicit equality-case or extremal examples, complementing Observation 7.1.

(4) **A proved instance of Observation 7.1.** The classification of short Bruhat intervals [12] is a plausible route to a genuinely proved (rather than empirical) equality classification, at least for intervals below some fixed length.

(5) E_8**.** No exhaustive or near-top data of any kind is recorded here for E_8 ($|W| > 696$ billion); even a thin near-top slab would extend Conjecture 5.1's evidence to every exceptional simply-laced type.

(6) **A quantitative form of Conjecture 2.11**, e.g. $\rho([u, v]) \geq 1 + c(W)$ with $c(W)$ explicit in simply-laced types decaying as predicted by Conjecture 5.1 and Theorem 6.1, would unify the extremal and asymptotic statements of this paper into a single sharp inequality.

We also note two points of context for the reliability of this line of work. First, log-concavity of Bruhat rank sequences is adjacent to, but distinct from, several actively studied log-concavity and equality-case questions — equivariant log-concavity of flag-variety cohomology [14], and equality cases of related inequalities [15, 16] — none of which, on inspection, concern the specific statement of Conjecture 2.11. Second, as a caution about treating any single open-problem survey as a stable ground truth: a different conjecture from the same survey we cite for Conjecture 2.11 was recently shown false [19]. A fresh literature sweep immediately preceding submission, repeating and extending the one performed when this project began, found nothing overlapping with any claim in this paper (see the reproducibility repository for the full query log).

Acknowledgments

This work was produced with substantial use of AI assistance, and we want to be specific about its scope rather than gesture at it in passing. Large language models (principally Claude, Anthropic,

in an agentic coding environment) wrote the great majority of the verification code, drafted proof arguments for the results in Sections 5 and 6, and assisted in assembling this manuscript from our working notes. Distinct AI sessions were used adversarially against each other's work where the project's own house rules required it: draft proofs were checked by independent review passes (including a from-scratch numerical audit) before we treated them as reliable, and at least three instances where an AI-drafted claim was numerically wrong on first pass — a sign error, a false finite-range certificate, and a fabricated verification result — were caught this way and corrected, not merely trusted.

None of this substitutes for human responsibility. The authors directed the research questions, judged which computational and proof strategies were worth pursuing, decided what could honestly be called a theorem versus what had to remain a conjecture, independently checked the disclosed constants and counterexamples, and take full responsibility for every claim in this paper, including any error that remains. We disclose the AI involvement because we think readers are entitled to know it, not to diffuse authorship: the choices of what to claim, what to hold back, and what still needs proof are ours.

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